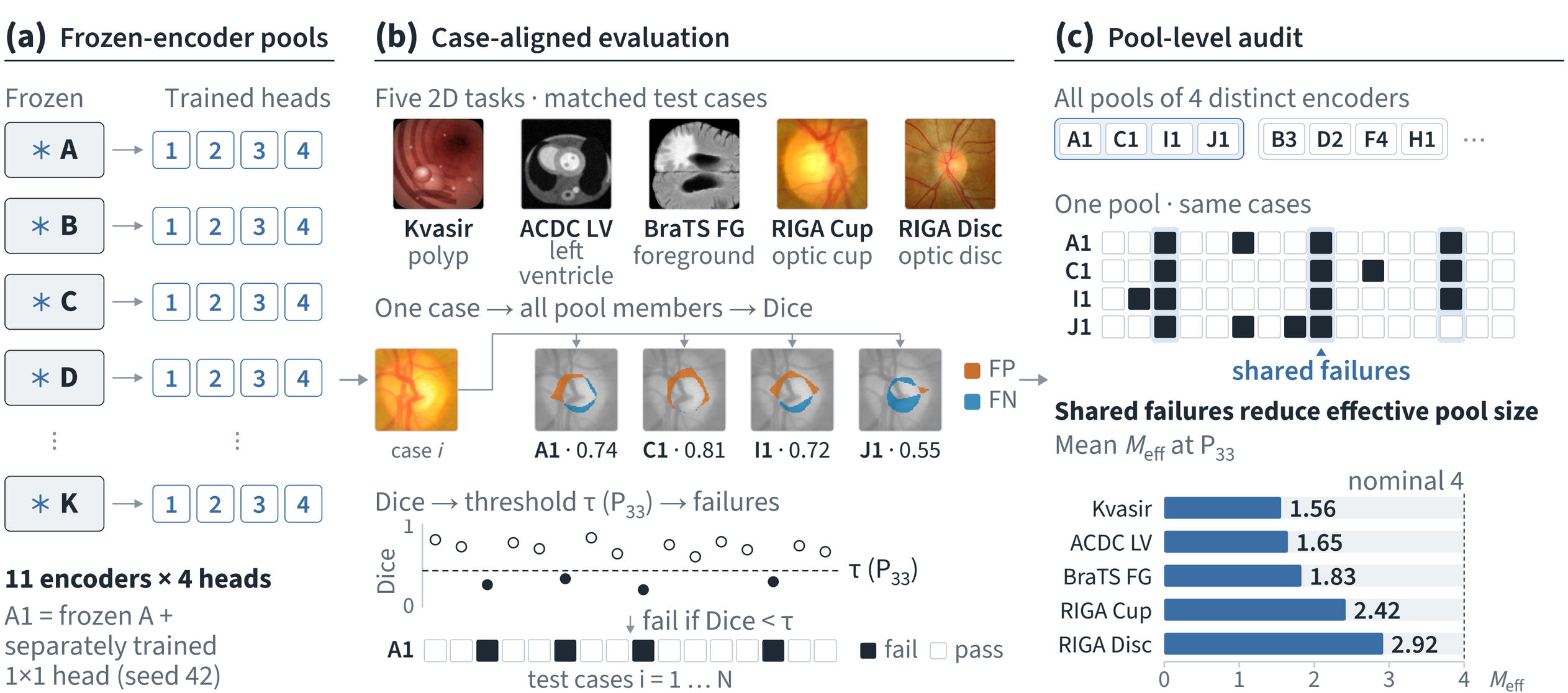

(a) Frozen-encoder pools
Frozen
Trained heads
∗ A
∗ B
∗ C
∗ D
∗ K
1 2 3 4
11 encoders × 4 heads
A1 = frozen A + separately trained 1×1 head (seed 42)
(b) Case-aligned evaluation
Five 2D tasks · matched test cases
Kvasir polyp
ACDC LV left ventricle
BraTS FG foreground
RIGA Cup optic cup
RIGA Disc optic disc
One case ⟶ all pool members ⟶ Dice
case i
A1 · 0.74
C1 · 0.81
I1 · 0.72
J1 · 0.55
FP
FN
Dice ⟶ threshold τ (P33) ⟶ failures
Dice
1
0
τ (P33)
↓ fail if Dice < τ
A1
fail
pass
test cases i = 1 … N
(c) Pool-level audit
All pools of 4 distinct encoders
A1 C1 I1 J1
B3 D2 F4 H1
…
One pool · same cases
A1
C1
I1
J1
shared failures
Shared failures reduce effective pool size
Mean M_{eff} at P_{33}
nominal 4
Kvasir 1.56
ACDC LV 1.65
BraTS FG 1.83
RIGA Cup 2.42
RIGA Disc 2.92
0 1 2 3 4
M_{eff}